Understanding Representation Bottlenecks in Neural Image-to-Text Models through Cross-Architecture Analysis

Rohan Anthony **Siyuan Wang** **Trisha Sharma** **Cirenyangzhen**
University of Bristol
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Abstract

Image captioning, the task of mapping visual input to natural language, is difficult to analyse in real-world settings where multiple factors interact. We study this problem in a controlled synthetic setting using geometric scenes with a closed vocabulary and deterministic image-to-caption mapping, allowing failures to be attributed to specific design choices. We evaluate four architecturally distinct approaches: a parallel classification model, a spatial attention LSTM, a Transformer decoder with patch-level cross-attention, and a controlled comparison of text representations. Across these, we independently vary encoder training, spatial feature structure, decoder architecture, and text representation. Fine-tuning the encoder emerges as the dominant factor, improving exact match from 51% to 99.73% in the parallel classification model and from 47% to 88% in the Transformer. Preserving spatial structure and applying attention produces an 82 percentage point gain (15% to 97%) over global pooling. In contrast, text representation has limited impact: pretrained GloVe embeddings underperform randomly initialised embeddings, and extended training improves performance only to 19.5% under a frozen backbone. A zero-shot evaluation of GPT-4o mini achieves 5.4% exact match, far below task-specific models. Despite their architectural differences, all approaches converge on the same conclusion: performance is primarily determined by the quality and structure of visual representations, while decoder sophistication plays a secondary role once this bottleneck is addressed.

1 Introduction

Image captioning, the task of mapping visual input to a natural language description, is a fundamental challenge in AI. However, given the complexity of real-world images, it is often difficult to understand why certain approaches perform better than others, since too many variables change for individual factors to be cleanly isolated.

To investigate this, we use a controlled synthetic setting, training multiple architectures on simple geometric scenes to isolate specific failure modes. The scenes generated are simple geometric shapes described by compositional attributes - shape, color, size and spatial relation, producing captions of the form *"a big red cube is above a small blue sphere"*. Because the vocabulary is closed and the mapping from scene to caption is deterministic, failures are informative in a way that natural image datasets are not.

This raises a direct question: what actually determines performance on this task? To answer it, we implement four architecturally distinct approaches and vary the encoder training strategy, decoder architecture, spatial feature structure and text representation independently across controlled experiments to understand how each design decision affects performance.

The base dataset consists of 2,000 synthetically generated 128×128 images across three scene types: 500 single-object, 1,000 binary two-object, and 500 triple three-object scenes. Binary scenes are generated from a fixed vocabulary of sizes, colours, shapes and spatial relations, with a single sentence template of the form *"a [size] [colour] [shape] is [relation] a [size] [colour] [shape]"*. The dataset is split 80/10/10 into train, validation and test sets. Larger datasets generated using the same procedure are used in specific experiments and are described below.

A good solution must correctly predict all seven attributes simultaneously - two sizes, two colours, two shapes, and one spatial relation across the full space of possible scenes. Exact match is the natural metric: a single wrong attribute fails the whole caption. Size discrimination presents a particular challenge, as marker-based rendering at 128×128 produces limited visual separability between size categories.

Our contributions are as follows:

- A synthetic dataset of 2,000 labelled scenes across three complexity levels, with a controlled vocabulary supporting causal analysis of failure modes.
- A five-stage ablation on a parallel classification architecture separating the effects of training data scale and encoder fine-tuning.
- A spatial attention LSTM isolating the contribution of preserved spatial feature maps to relational reasoning.
- A Transformer decoder with patch-level cross-attention providing a third independent test of encoder adaptation effects.
- A controlled comparison of text representations: random embeddings, one-hot encoding, and GloVe vectors.
- A zero-shot evaluation of GPT-4o mini as an external calibration point for task difficulty.

Across all four approaches, performance bottlenecks consistently originate in the visual representation stage. Encoder adaptation, spatial feature preservation, and representation quality determine performance. Decoder sophistication has limited marginal impact once this bottleneck is addressed. We show how four architecturally distinct approaches arrive at this conclusion through four different failure modes.

2 Methods

2.1 Dataset and Scene Generation

The base dataset consists of synthetically generated scenes composed of simple geometric objects. Each object is defined by a discrete set of attributes: three sizes, eight colours, eight shapes, and eight spatial relations. Binary scenes follow a fixed sentence template of the form *"a [size] [colour] [shape] is [relation] a [size] [colour] [shape]"*. This design ensures a closed vocabulary and a deterministic mapping between image and caption, meaning that any prediction error can be attributed to a specific attribute rather than ambiguity in the data. While this simplifies the task, it also introduces a limitation: all captions share the same structure, so models are not required to learn variation in sentence composition.

To support different experimental objectives, extended versions of the dataset are used in specific settings. For the encoder ablation, the binary dataset is expanded to approximately 12,000 samples to isolate the effect of data scale under both frozen and fine-tuned backbones. Separately, a revised dataset generation pipeline introduces greater variation in caption structure and visual appearance. This version incorporates randomised background colours to increase visual variation. Duplicate scenes are removed, allowing generation of up to 15,000 unique samples. These extensions are used to test whether increased data diversity or visual fidelity changes the observed performance trends, and are referenced in the relevant experimental sections.

A known challenge in the dataset arises from size discrimination. All images are rendered at a resolution of 128×128 using marker-based plotting, which limits the visual separation between size categories. As a result, differences between "small", "medium", and "large" objects can be subtle, particularly under spatial overlap or noise. This constraint is inherent to the rendering process and represents a limitation of the dataset, not necessarily a failure of the models themselves.

2.2 Baseline: Parallel Classification Architecture

The baseline model is designed around a simple question: should caption generation be treated as a sequential prediction problem, or as a set of independent attribute predictions? In this task, each caption encodes seven attributes: two sizes, two colours, two shapes, and one spatial relation. These attributes are structurally parallel rather than sequential, suggesting that a decoder that predicts them independently may be better aligned with the task than one that generates tokens step-by-step. The hypothesis is that removing unnecessary sequential dependence will simplify learning and improve accuracy. The limitation with this architecture is that it can only be applied to binary object scenes without dynamically adjusting the number of parallel classification heads.

The model uses a ResNet18 encoder He et al. [2016] to extract a global visual representation, followed by seven independent classification heads. Each head corresponds to one attribute: size, colour, and shape for each of the two objects, and a single head for the spatial relation. Each head is implemented as a two-layer MLP consisting of a linear projection (256→128), followed by ReLU activation and dropout (0.3), and a final linear layer mapping to the corresponding vocabulary. This design directly maps the structured output space to the model architecture.

Training is performed under two conditions: with the encoder frozen and with the encoder fine-tuned. This distinction forms the primary axis of the ablation study described in the next section. All other components of the model are held constant, allowing the effect of encoder adaptation to be isolated.

2.3 Axis 1: Encoder Fine-Tuning: Five-Stage Ablation

The first experimental axis examines the role of encoder adaptation through a five-stage ablation. The design varies two factors independently: dataset size and whether the backbone is frozen. The key question is whether the performance plateau observed under a frozen encoder is due to insufficient data, or whether it reflects a deeper limitation in using ImageNet-pretrained features for synthetic geometric scenes.

Across the five stages, the ablation separates the impact of data scale and encoder adaptation. Early stages test increasing dataset size under a frozen backbone, while Stage 5 introduces end-to-end fine-tuning with the full large dataset. This setup allows the contribution of each factor to be evaluated independently, ensuring that any observed performance changes can be attributed to a specific design choice.

2.4 Axis 2: Spatial Feature Structure: Attention LSTM

The second axis focuses on how spatial information is represented and accessed during decoding. This model uses a ResNet18 encoder that preserves spatial structure by outputting a 7×7 feature map. The central hypothesis is that relational reasoning requires access to spatially structured features, and that collapsing these features into a single global vector removes information necessary to predict spatial relations such as “above” or “to the left of”.

The decoder is an LSTM equipped with an additive attention mechanism over the 49 spatial positions. At each decoding step, attention weights are computed over the feature map, allowing the model to focus on different regions of the image when generating different parts of the caption. This provides a dynamic view of the scene.

To isolate the effect of spatial structure, this model is compared against a baseline LSTM that operates on a globally pooled feature vector. This comparison holds the decoder type constant while removing spatial information, allowing the contribution of attention over structured features to be evaluated directly.

2.5 Axis 3: Decoder Architecture: Transformer with Patch Cross-Attention

The third axis explores an alternative decoding paradigm using a Transformer decoder with cross-attention over spatial features. The model attends to 49 spatial patches at each decoding layer, without relying on a recurrent hidden state. The hypothesis is that patch-level cross-attention provides richer and more direct access to visual information at each step of generation.

A practical challenge in this setup is computational cost. To address this, features from the encoder are precomputed and cached when the backbone is frozen. This significantly reduces training time and makes large-scale experiments feasible on limited hardware. This design choice enables a direct comparison between frozen and unfrozen backbones without requiring repeated forward passes through the encoder.

As with the baseline model, the primary variable of interest is whether the encoder is frozen or fine-tuned. The Transformer architecture serves as a third independent test of the role of encoder adaptation, as opposed to a direct competitor to the other decoders.

2.6 Axis 4: Text Representation

The final axis examines the role of text representation in the task. Three alternatives are compared using a separately implemented model with a frozen MobileNetV2 backbone Sandler et al. [2018]: randomly initialised embeddings, one-hot encoding with linear projection, and pretrained GloVe vectors Pennington et al. [2014]. This axis is motivated by the fact that, although the task is visually grounded, the decoder must still learn a mapping from visual features to token sequences, and different token representations may affect how easily this mapping is learned.

We hypothesise that random embeddings may already be sufficient in this small, highly regular vocabulary setting. One-hot representations may suit the discrete attribute structure of the task, while GloVe may offer limited benefit because the synthetic captions do not reflect natural language co-occurrence patterns.

This axis tests whether language-side improvements can compensate for limitations in the visual encoder. If performance remains limited across text representations, this suggests that the dominant bottleneck lies in the visual representation rather than in the decoder or token encoding. The comparison therefore serves as a controlled test of how far language-side modifications alone can improve performance in this task.

2.7 LLM Comparison: GPT-4o mini

A zero-shot evaluation using GPT-4o mini OpenAI [2024] is included as an external reference point. The goal is not to compare directly against a task-specific model, but to assess the relative difficulty of the task and the value of supervised training. A subset of 130 images is evaluated using a fixed prompt, and outputs are parsed into the same structured format used for model evaluation.

This comparison provides context for the results: if a general-purpose model performs poorly, it suggests that the task requires precise grounding in visual features that cannot be recovered from language priors alone. Conversely, strong performance would indicate that the task is largely solvable through general reasoning.

3 Experimental Evaluation

3.1 Metrics and Their Limitations

Exact match is used as the primary evaluation metric throughout all experiments. A prediction is counted as correct only if all seven attribute slots: size_1, colour_1, shape_1, relation, size_2, colour_2, shape_2 are predicted correctly simultaneously. This is a deliberate choice given the task structure: captions are compositional and deterministic, so a prediction that recovers six of seven attributes but misidentifies, for example, a spatial relation produces a caption that is factually wrong. Partial credit would obscure this. The harshness of exact match is therefore a feature rather than a limitation in this setting.

Per-slot accuracy is used as a supplementary diagnostic. Because exact match conflates all failure modes, per-slot accuracy allows errors to be attributed to specific attributes and compared across architectures. This decomposition is particularly useful for identifying systematic weaknesses, for instance, whether size errors are more common than relation errors, or whether a particular architecture struggles with a specific slot.

Sequence-level metrics such as BLEU are not used. BLEU measures n-gram overlap between generated and reference strings, which is appropriate when outputs vary in length and phrasing. In this task, all captions share a fixed seven-attribute template, so n-gram overlap between a wrong and correct caption can be high even when multiple attributes are wrong. BLEU would reward partially correct predictions in a way that misrepresents model performance on a closed-vocabulary compositional task.

3.2 Experimental Setup

All experiments use the same 80/10/10 train/validation/test split. The base dataset of 2,000 samples provides test sets of approximately 100 binary scenes. Extended datasets used by the encoder ablation and Transformer experiments are generated using the same rendering pipeline with larger binary sample counts; in these cases the test set scales proportionally, yielding approximately 1,494 binary test scenes for the 16,000-sample dataset. Where extended datasets are used, this is noted explicitly in the relevant section.

The encoder ablation (Section 3.3) trains for 30 epochs in Stages 1-3, 50 epochs in Stage 4, and runs until convergence in Stage 5. The Adam optimiser is used with a learning rate of 10^{-3} for classification heads. When the backbone is unfrozen, a differential learning rate of 10^{-4} is applied to backbone parameters. Batch size is 64 throughout. The spatial attention LSTM and Transformer decoder experiments use Adam with learning rates in the range 10^{-3} to 10^{-4} as determined per experiment. Text representation comparisons are evaluated at 3 and 30 epochs under a frozen backbone. All experiments are implemented in PyTorch. Pretrained ResNet18 weights are loaded via `torchvision.models`; MobileNet weights are loaded from the same source. GPT-4o mini is accessed via the OpenAI API with a fixed structured prompt.

3.3 Results: Encoder Fine-Tuning Ablation

The plateau under frozen features. Stages 1 through 4 hold the backbone frozen while increasing the number of binary training samples from 801 to 12,012 and extending training from 30 to 50 epochs. Table 1 reports exact match accuracy across all five stages. Performance improves with data from 13.13% at 801 samples to 48.53% at 12,012 samples but plateaus well below ceiling. Extending training from 30 to 50 epochs at the same data scale (Stages 3 and 4) yields only a marginal gain of 2.5 percentage points (48.53% to 51.00%), confirming that the model had converged rather than simply undertrained. The plateau persists regardless of data scale, establishing that frozen ImageNet features are the binding constraint, not the amount of training data.

Table 1: Five-stage encoder ablation. Stage 1 is evaluated on approximately 99 binary test scenes, Stage 2 on approximately 499, and Stages 3–5 on 1,494 scenes, corresponding to the 80/10/10 split of their respective datasets.

Stage	Train samples	Backbone	Epochs	Exact match
1	801	frozen	30	13.13%
2	4,007	frozen	30	25.65%
3	12,012	frozen	30	48.53%
4	12,012	frozen	50	51.00%
5	12,012	fine-tuned	6	**99.73%**

The fine-tuning breakthrough. Stage 5 introduces end-to-end fine-tuning with a differential learning rate: backbone parameters are updated at $0.1\times$ the head learning rate to prevent destructive interference with pretrained representations. The model reaches 99.73% exact match (1,490 of 1,494 test scenes correct) and converges in six epochs. The speed of convergence is itself informative: the model does not require large numbers of gradient steps once the backbone is free to adapt. The conclusion is that the bottleneck across Stages 1–4 was encoder adaptation, not data volume or decoder capacity. Frozen ImageNet features, trained on natural photographic images, cannot adequately represent the flat-colour synthetic geometry of the rendered scenes. Fine-tuning removes this constraint entirely.

Residual errors and size slot concentration. The four remaining failures at Stage 5 are attributable to size misclassification in three cases (size$_1$ predicted incorrectly in two instances, size$_2$ in one) and a single relation error (lower-left-of predicted as above). Per-slot accuracy at Stage 5 is reported in Table 2. Colour, shape, and spatial relation slots all achieve at or near 100%; size slots are the sole source of systematic error. This pattern is consistent across all five stages: $size_1$ is the weakest slot at every stage of the ablation. The attribution is not model capacity but rendering resolution. At 128×128, visual separability between size categories is limited, and the difficulty is inherent to the input signal rather than addressable by architectural changes.

Table 2: Per-slot accuracy at Stage 5 (fine-tuned, 12,012 training samples). Test set: 1,494 binary scenes.

Slot	Correct / Total	Accuracy
$size_1$	1,492 / 1,494	99.87%
$colour_1$	1,494 / 1,494	100.00%
$shape_1$	1,494 / 1,494	100.00%
relation	1,493 / 1,494	99.93%
$size_2$	1,493 / 1,494	99.93%
$colour_2$	1,494 / 1,494	100.00%
$shape_2$	1,494 / 1,494	100.00%

3.4 Results: Spatial Attention LSTM

Global pooling as the within-experiment control. The spatial attention experiment establishes its own baseline by first evaluating a plain LSTM decoder operating on a globally pooled feature vector from a ResNet18 encoder, with the backbone unfrozen throughout. This configuration achieves 15% exact match. Because the backbone is unfrozen in both conditions, the comparison isolates the effect of spatial feature structure on decoder performance while holding encoder adaptation constant.

Spatial attention over preserved feature maps. Replacing the global pooled vector with attention over the 7×7 spatial feature map raises exact match accuracy to 97%. The gain of 82 percentage points from a single architectural change: preserving spatial structure in the encoder output and equipping the decoder with an attention mechanism over it is large and clean. The result supports the hypothesis that relational reasoning between two objects requires the decoder to access spatially differentiated information about image regions. Collapsing spatial structure into a single vector removes the information needed to accurately predict attributes such as "above" or "to the left of". Beam search decoding on the global pooling baseline improves greedy exact match only from 15% to 21%, confirming that the bottleneck lies in the model representation rather than the decoding strategy. Table 3 summarises the Group A results.

Table 3: Group A: core architecture comparison (ResNet18 encoder, all runs fine-tuned end-to-end).

Run	Setup	Val Loss	Greedy Exact	Beam Exact
A1	LSTM (global pool)	0.4728	0.15	0.21
A2	LSTM + CBAM	0.4592	0.17	0.22
A3	**Attention-LSTM**	**0.0127**	**0.97**	**0.97**

Convergence with the encoder ablation finding. The two experiments identify different failure modes but the same underlying diagnosis. In the encoder ablation, frozen ImageNet features prevent the representation from capturing the synthetic domain; in the spatial attention experiment, collapsing spatial structure removes information the decoder needs for relational reasoning. In both cases, the failure originates in the visual representation before the decoder, not in the decoder itself. This convergence across independent experiments strengthens the central claim: getting the visual representation right is the prerequisite for either approach to work.

3.5 Results: Transformer Decoder

Frozen versus unfrozen backbone. The Transformer decoder experiment tests a third decoding paradigm: cross-attention over 49 spatial patches at each decoding layer, with no recurrent hidden state. Under a frozen backbone, exact match is 47%. With end-to-end fine-tuning, this rises to 88%. The pattern is consistent with both previous experiments: backbone adaptation is the primary driver of performance, and the frozen-to-fine-tuned gap is large regardless of decoder architecture. Training under a frozen backbone completed 30 epochs in under 5 minutes (10 seconds per epoch), but produced systematic colour confusion, most notably "cyan" predicted as "blue", which is consistent with frozen ImageNet features being unable to distinguish perceptually similar flat synthetic colours. With the backbone unfrozen, training took approximately 85 seconds per epoch over 50 epochs, and colour confusions are eliminated entirely, confirming that backbone adaptation is the decisive factor in resolving fine-grained colour discrimination. Table 4 reports results for both configurations.

Table 4: Transformer decoder results on 1,600 training samples. Backbone fine-tuning is the primary variable.

Configuration	Backbone	Exact Match	Token Accuracy
Frozen backbone	frozen	47.00%	88.80%
Unfrozen backbone	**fine-tuned**	**88.00%**	**98.41%**

The gap from 88% to ceiling. The fine-tuned Transformer reaches 88%, below the 97% and 99.73% ceilings achieved by the spatial attention LSTM and parallel classification model respectively. This gap is attributable to dataset scale rather than decoder architecture. The Transformer experiment uses the 2,000-sample base dataset; the other two experiments use extended datasets of 12,000–15,000+ binary training samples. The relationship between data scale and performance observed in the encoder ablation (Section 3.3) predicts that a fine-tuned Transformer trained on the larger dataset would approach a similar ceiling. The 88% result should therefore not be interpreted as a decoder failure but as a data scale effect consistent with the broader experimental pattern.

A third independent confirmation. The Transformer decoder is architecturally distinct from both the parallel classification model and the spatial attention LSTM: it uses neither parallel output heads nor a recurrent hidden state, instead relying on self-attention and cross-attention mechanisms. All three approaches exhibit the same backbone-dependence pattern: large gains from fine-tuning and only modest gains from decoder sophistication once backbone adaptation is controlled. Each experiment reaches this conclusion through a different architectural path.

3.6 Results: Text Representation

Ordering of text representations. Three text representation strategies are evaluated under a frozen MobileNet backbone: randomly initialised embeddings, one-hot encoding, and pretrained GloVe vectors. Table 5 reports exact match and token accuracy across all three representations. At 3 epochs, exact match accuracy is 6.0% for random embeddings, 2.5% for one-hot, and 1.0% for GloVe. Random embeddings also achieve the highest token accuracy at 62.64%. The ordering suggests that pretrained or more structured language representations do not automatically improve performance in this setting.

Table 5: Text representation comparison (MobileNetV2 encoder, frozen, 3 epochs). Exact match on binary test scenes.

Representation	Exact Match	Token Accuracy
Random embeddings	**6.0%**	**62.64%**
One-hot + projection	2.5%	47.36%
GloVe (100d)	1.0%	55.68%

Why richer representations do not help. GloVe vectors are learned from natural language corpora, where words are organised by distributional co-occurrence. In this task, however, the vocabulary is small, closed, and synthetic, and the captions follow fixed compositional templates. The semantic structure encoded by GloVe therefore does not transfer cleanly. Random embeddings, by contrast, are free to organise the vocabulary according to the task-specific distinctions required by the visual prediction problem. One-hot representations preserve the discrete attribute structure of the task, but appear less efficient than a learned continuous embedding space.

Extended training. To test whether the 3-epoch comparison reflects convergence or simply undertraining, random embeddings were trained for 30 epochs with learning-rate decay. Exact match improved to 19.5%, with the best checkpoint at epoch 26. This shows that the low 3-epoch result partly reflects an undertrained decoder rather than a hard ceiling imposed by the representation itself.

The binding constraint. Even after extended training, the best text representation remains far below the performance achieved by models with stronger visual representations in the other experimental axes. This experiment therefore acts as a control: language-side improvements affect learning efficiency, but they do not compensate for limitations in the visual encoder. The dominant bottleneck remains on the visual side of the pipeline. Critically, this reinforces the broader finding that once the visual representation is constrained, improvements on the decoder or output representation side cannot recover the missing information.

3.7 LLM Comparison: GPT-4o mini

A zero-shot evaluation of GPT-4o mini is conducted on a sample of 130 images using a fixed structured prompt. The model is asked to describe each scene in the same seven-attribute format used throughout this work; outputs are parsed into slot predictions and evaluated against ground truth using the same exact match criterion. GPT-4o mini achieves an exact match accuracy of 5.4% (7/130 scenes) on this sample. Per-slot analysis reveals colour identification is strongest at 85.4%, shape at 73.1%, and size at 26.9%, below the 33.3% expected from random guessing across three categories, suggesting the model performs worse than chance on size discrimination. Nine responses contain out-of-vocabulary terms (e.g. “square” for “cube”), indicating the model’s vocabulary is not aligned with the closed synthetic vocabulary used in this task.

The result confirms that supervised training on domain-specific data provides a decisive advantage. GPT-4o mini’s 5.4% exact match, compared to 88%–99.73% achieved by fine-tuned task-specific models, demonstrates that zero-shot visual reasoning cannot substitute for targeted encoder adaptation on this task. Colour and shape identification generalise reasonably well (85.4% and 73.1%), but size discrimination, which is the same attribute that causes residual errors in fine-tuned models, is the primary bottleneck at 26.9%, consistent with the rendering resolution constraint identified across all experimental axes.

3.8 Convergent Error Analysis

Size as a universal failure mode. Across all four experimental axes, size prediction is the most error-prone attribute. In the encoder ablation, size_1 is the weakest slot at every stage, and three of the four remaining failures at 99.73% are size errors. In the Transformer and spatial attention experiments, size slots similarly account for a disproportionate share of per-slot errors. A second cross-architecture failure mode under frozen backbones is colour confusion: the Transformer experiment shows systematic misclassifications of perceptually similar colours (e.g. “cyan” predicted as “blue”) that disappears entirely once the backbone is fine-tuned, providing further evidence that frozen ImageNet features cannot adequately represent the flat synthetic colour palette of the rendered scenes. At 128×128 resolution, marker-based rendering produces limited visual separation between size categories; small differences in marker diameter must be resolved from a constrained pixel budget. No architectural intervention addresses this directly, and the size error rate does not substantially diminish until the backbone is fine-tuned end-to-end and the dataset is large enough to provide adequate examples of all size combinations.

Spatial relation errors and their conditions. Relation errors are rare in fine-tuned models: the spatial attention LSTM and parallel classification model both achieve above 99% relation accuracy, and all eight relation classes are predicted correctly by the fine-tuned parallel classifier except for a single instance of *lower left of* predicted as *above*. Relation errors are more frequent under frozen backbones and low data conditions. In the spatial attention experiment, the global pooling baseline (15% exact match) suggests that the precision of the relation suffers substantially when the spatial structure is collapsed, consistent with the hypothesis that spatial relations require spatially differentiated visual information.

4 Conclusions

Across all experiments, fine-tuning the encoder backbone is the single most impactful intervention. In the parallel classification model, it enables a jump from a plateau of 51% under frozen features to 99.73% exact match. The Transformer decoder exhibits the same pattern, improving from 47% to 88% with end-to-end training. Spatial attention over preserved feature maps provides a second, independent mechanism for improvement, raising performance from 15% to 97% within the LSTM framework. Although these approaches differ architecturally, both address the same underlying issue: ensuring that the visual representation contains the information required for the task.

In contrast, several interventions show limited impact. Increasing dataset size under a frozen backbone produces diminishing returns and fails to approach ceiling performance. Improvements in decoder architecture alone do not overcome poor visual representations. On the language side, pretrained GloVe embeddings underperform randomly initialised embeddings, reflecting a mismatch between natural language semantics and a closed synthetic vocabulary. A zero-shot evaluation of GPT-4o mini achieves only 5.4% exact match, demonstrating that general-purpose visual reasoning cannot substitute for task-specific training.

The most important result is the convergence of these findings. Four architecturally distinct approaches, namely parallel classification, spatial attention LSTM, Transformer decoding, and text representation control, arrive at the same diagnosis through different failure modes. In one case, the limitation is a frozen encoder, in another, collapsed spatial structure, in a third, insufficient data for adaptation, and in a fourth, ineffective output representations. Despite these differences, all point to the same conclusion: performance is determined by the quality and structure of visual representations before decoding begins.

The study has several limitations. Size discrimination remains imperfect even after fine-tuning, due to limited visual separability at 128×128 resolution. The base dataset uses a single sentence template, restricting linguistic variability. Triple-object scenes are not evaluated across all fine-tuned models, limiting conclusions about compositional generalisation. The parallel classification architecture is also constrained to binary scenes and does not extend naturally to variable numbers of objects.

More broadly, these results suggest that in image captioning, encoder pretraining and adaptation to the target domain are more critical than decoder sophistication. Synthetic, controlled datasets provide a valuable framework for isolating causal factors in model performance, enabling a level of interpretability that is difficult to achieve with natural images. The findings here should therefore be understood as a controlled demonstration of how representation quality governs downstream performance.

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